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Louis Fostier

Education and professional experiences

- 02/2026–
Present **Postdoctoral Researcher, Modeling of adipose tissue morphogenesis, ENERGENCE project**, team MUSCLEES, INRIA, Paris, France.
Keywords: Computational biology ; Individual-based model ; Sensibility analysis ; Metamodeling ; Tissue morphogenesis
- 03/2024–
05/2024 **Research stay**, *Department of mathematics*, Duke University, USA.
Dissemination of PhD research work, Mathematical modelling in biology, Biologically Informed Neural Networks methods
- 11/2022–
10/2025 **PhD student on Analysis and calibration of structured population dynamics models: application to cell populations**, team BIOS, INRAE, PRC, Nouzilly, France.
Mathematical modelling in biology, Well-posedness and long-time analysis of size-structured population dynamics models, numerical methods for inverse problem.
Keywords: Size-Structured Population Dynamics ; Asymptotic Behavior ; Bifurcation Analysis ; Inverse Problem ; PINNs ; Reproductive Biology
- Advisors: Frédérique Clément (team MUSCA, INRIA), Romain Yvinec (team BIOS, INRAE) and Violette Thermes (LPGP, INRAE).
- 2020–2022 **Master of Applied Mathematics**, *Université de Rennes 1*.
Scientific computing, modelling, numerical analysis. Obtained with very good honors.
- 2017–2020 **Bachelor of Mathematics**, *Université de Rennes 1*.
Obtained with very good honors.

Internship and Hackaton

- 2023 **6 weeks hackaton on Scientific machine learning for equation discovery**, *Centre d'Été Mathématique de Recherche Avancée en Calcul Scientifique*, Marseille, France.
- 2022 **M2 internship on Modelling of the female fish germ cell population**, team MUSCA, INRIA, Palaiseau, France.
- Advisors: Frédérique Clément (team MUSCA, INRIA) and Romain Yvinec (team BIOS, INRAE)
- 2021 **M1 internship on Scientific computing and modelling for biofilms**, IRMAR, CNRS, Rennes, France.
- Advisors: Fabrice Mahé (IRMAR, CNRS) and Bénédicte Martin (NUMECAN, INSERM).

Publications

Journal Article

- 2026 **Fostier L., Dauger A., Yvinec R., Ribot M., Audebert C., Soula H.**, Rapid cell turnover to model adipocyte size distribution, in *Journal of Theoretical Biology*.
<https://doi.org/10.1016/j.jtbi.2025.112311>
- 2025 **Clement F., Fostier L., Yvinec R.**, Bifurcation analysis of a size-structured population model: Application to oocyte dynamics and ovarian cycle, in *SIAM Journal on Applied Dynamical Systems (SIADS)* (Selected by scientific committee for ReproHighlights GdR Repro 2025) .
<https://doi.org/10.1137/24M1705147>

- 2022 **Chathoth K., Fostier L., Martin B., Baysse C., Mahé F.**, A Multi-Skilled Mathematical Model of Bacterial Attachment in Initiation of Biofilms, in *microorganisms*.
<https://doi.org/10.3390/microorganisms10040686>

Proceeding

- 2025 **Fiorini C., Flint C., Fostier L., Franck E., Hashemi R., Michel-Dansac V., Tenachi W.**, Generalizing the SINDy approach with nested neural networks, in *ESAIM: Proceedings and surveys*.
<https://doi.org/10.1051/proc/202581168>

Preprint

- 2026 **Fostier L., Lesage M., Thermes V., Clément F., Yvinec R.**, Data-driven PINN inference of oocyte dynamics in fish ovaries.
<https://inria.hal.science/hal-05536280>

Communications in conference and seminar

Oral communications

- 2026 Workshop on Data-driven Approaches in Modelling Cell and Tissue Dynamics, October 14-16, LJLL Paris
 2026 Metaprogramme INRAE DIGIT-BIO seminar, September 16-18, Montpellier
 2026 WPI Workshop on Mathematical modeling in biology and medicine, July 22-25, Wien Austria
 2026 European Conference on Mathematical and Theoretical Biology, July 13-17, Graz, Austria
 2026 Conference Data assimilation and inverse problems for biomedical problems, June 16, Nantes
 2026 CANUM 2026 de la SMAI, June 3, Saint Jacut
 2025 Math seminar of the Laboratoire de Mathématiques Appliquées du Havre, November 27, Le Havre
 2025 Math Bio seminar of Institut Denis Poisson, September 24, Tours
 2025 Differential Equations and Applications to Biology, June 16-20, Le Havre
 2025 Biennale de la SMAI, June 02-05, Carcans-Maubuisson
 2024 Métaprogramme DIGIT-BIO seminar, December 12-13, Lyon
 2024 Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées, October 28-30, Lyon
 2024 Biology and Medicine Through Mathematics Conference, May 15-17, Richmond USA
 2024 Math bio seminar, March 22, Duke University USA
 2023 Journées Math-Bio-Santé, November 27 - December 1, Paris Marne La Vallée
 2023 Math Bio seminar of Institut Denis Poisson, July 13, Orléans
 2023-2025 PhD student seminar of the research unit Physiologie de la Reproduction et des Comportements
 2023-2024 PhD student seminar week of Institut Denis Poisson

Posters

- 2026 Physics-Informed Machine Learning Day, April 14-15, Agro Paris Saclay
 2025 Journées Math-Bio-Santé, November 05-07, Montpellier
 2024 European Conference on Mathematical and Theoretical Biology, July 22-26, Toledo
 2024 Journées Math-Bio-Santé, June 24-28, Nantes
 2023 Journées reprosciences, 3-5 May, Paris

Teaching and tutoring

- 2024 **L1 classes of Calculus and mathematical reasoning**, Université de Tours, France.
 30 hours
 2022 **L1 classes of Algebra and Analysis**, Université de Tours, France.
 54 hours
 2021 **Assistance and pedagogical support for a disabled student**, Université de Rennes 1, France.
 2020 **Educational tutoring for students in L3 mathematics**, Université de Rennes 1, France.
 2025-26 **Tutoring for elementary and middle school students**, Maison de la réussite (Tours) and Espace 19 (Paris), France.

Popularisation

2024 **IA workshop**, *Seminar of the INRAE research unit Physiologie de la Reproduction et des Comportements*.

Research stays abroad

2024 **Duke University**, *March 11 – May 10, 2024*, INRIA funding (€10k) – Sabbatical program: Dissemination of PhD research work, support from Kevin Flores (NC State University) to explore BINN (Biologically Informed Neural Network) methods.

Awards

2024 Best poster award - European Conference on Mathematical and Theoretical Biology, Toledo Spain

Computer skills

Basic C++

Intermediate C, FORTRAN, MATLAB, FREEFEM++

Advanced PYTHON

Languages

Fluent English

Advanced Italian

Basic Spanish

Beginner Russian

Referees

Dr. Frédérique Clément

Research Director, head of MUSCA team
Inria Saclay

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Dr. Magali Ribot

University professor
Institut Denis Poisson

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Dr. Romain Yvinec

INRAE Research Director, head of BIOS team
INRAE Val-de-Loire

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